Lecture notes. Lecture 10

Reinforcement learning
Concise companion to the full lecture notes

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Reinforcement learning, its value functions and its learning algorithms come first in these notes. Quantum state preparation then serves as the implementation example, introduced through its goal, primitive operations and target states. The notation and the derivation of the value functions follow the reinforcement-learning chapter of Ref. [1,2].

1 Agent, environment and reward

An agent observes a state s_t, chooses an action a_t, and receives the next state s_{t+1} and a scalar reward r_{t+1}. A policy $\pi(a \mid s)$ gives the probability of choosing an action in a state. One episode is a sequence of these transitions ending at a terminal state. The reward specifies the goal, and the agent must discover which choices lead to a large total reward.

The state must contain the information needed to predict the next outcome from the current action, which is the Markov property. If a remaining time or action budget affects the continuation, it is part of that state. The environment model is

$$p(s', r \mid s, a) = \Pr(s_{t+1} = s', r_{t+1} = r \mid s_t = s, a_t = a).$$

An agent with access to this model can plan with it, whereas model-free learning uses observed transitions. The policy and the environment can both introduce randomness.

The discounted return from step t is

$$G_t = \sum_{j=0}^{T-t-1} \gamma^j r_{t+j+1} = r_{t+1} + \gamma G_{t+1}, \qquad 0 \le \gamma \le 1. \tag{1.1}$$

Here T is the final step, and γ controls the weight of rewards received later. The objective is the expected return. Exploration tries uncertain actions to gather information, and exploitation chooses the action that currently appears best. Their balance matters because a later reward can depend on several earlier choices.

2 Values of states and actions

We fix a policy π. Its state value is the mean return when the agent starts at s and follows that policy,

$$V_\pi(s) = \mathbb{E}_\pi[G_t \mid s_t = s]. \tag{2.1}$$

It is the expected return from continuing the episode at the current state. It depends on the policy, because the later actions affect the outcome.

The action value fixes the next action before following the same policy,

$$Q_\pi(s, a) = \mathbb{E}_\pi[G_t \mid s_t = s, a_t = a]. \tag{2.2}$$

It is the expected return from taking a particular action now and then continuing with π. Averaging over that action with the policy probabilities gives the state value,

$$V_\pi(s) = \sum_a \pi(a \mid s) Q_\pi(s, a). \tag{2.3}$$

The advantage compares the action with this policy average,

$$A_\pi(s, a) = Q_\pi(s, a) - V_\pi(s). \tag{2.4}$$

Positive advantage means that choosing a gives a larger expected return than letting the policy choose in the same state. The comparison concerns the whole remaining return, including later actions.

2.1 The Bellman equation

We insert the recursion of the return from equation (1.1) and average over the policy's next action and over the state and reward that the environment returns,

$$\begin{aligned} V_\pi(s) &= \mathbb{E}_\pi[r_{t+1} + \gamma G_{t+1} \mid s_t = s] && (2.5)\\ &= \sum_a \underbrace{\pi(a \mid s)}_{\text{choice of action}} \sum_{s',r} \underbrace{p(s', r \mid s, a)}_{\text{outcome of the action}} \quad [r + \gamma \mathbb{E}_\pi[G_{t+1} \mid s_{t+1} = s']] && (2.6)\\ &= \sum_a \pi(a \mid s) \sum_{s',r} p(s', r \mid s, a)[r + \gamma V_\pi(s')]. && (2.7) \end{aligned}$$

The Markov property makes the continuation depend on s' alone, and since all later actions follow π, the inner expectation is $V_\pi(s')$. Fixing the first action gives

$$Q_\pi(s, a) = \sum_{s',r} p(s', r \mid s, a)[r + \gamma V_\pi(s')]. \tag{2.8}$$

A terminal state has zero continuation value because no further rewards follow it.

The optimal value V_* is the largest expected return over policies. Its next action is the best one,

$$V_*(s) = \max_a \sum_{s',r} p(s', r \mid s, a)[r + \gamma V_*(s')]. \tag{2.9}$$

The corresponding action value obeys

$$Q_*(s, a) = \sum_{s',r} p(s', r \mid s, a)[r + \gamma \max_{a'} Q_*(s', a')]. \tag{2.10}$$

Choosing $\arg\max_a Q_*(s, a)$ is optimal for the specified return. When the transition model is known, value iteration repeatedly applies the right-hand side to a current estimate. For bounded rewards and $\gamma < 1$ this operation contracts the largest error by at least a factor γ. A finite-horizon problem is solved backwards from its terminal values. These methods use the model, whereas Q-learning uses observed transitions.

3 Learning a table from experience

For a finite collection of states and actions, a Q-table stores one estimate in each cell, with states labelling its rows and actions its columns. A transition updates the cell of the state and action that were visited. A literal table applies to a finite state and action space or a chosen discretisation. A function approximator is needed when a table would be too large or the state space is continuous.

To estimate the state values of a fixed policy, we move the estimate towards a target,

$$V(s_t) \leftarrow V(s_t) + \alpha[\text{target} - V(s_t)], \qquad 0 < \alpha \le 1. \tag{3.1}$$

Monte Carlo uses the observed return G_t after the episode ends. Temporal-difference (TD) learning uses $r_{t+1} + \gamma V(s_{t+1})$ after one transition. Its correction is

$$\delta_t = r_{t+1} + \gamma V(s_{t+1}) - V(s_t). \tag{3.2}$$

The next reward is observed, and the rest is estimated from the next state. Using a value estimate in the target is called bootstrapping. A terminal continuation is zero.

3.1 Q-learning

Q-learning estimates the optimal action values by a sampled version of equation (2.10),

$$Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]. \tag{3.3}$$

The update compares the existing cell with the observed reward plus the best estimated continuation. If a table cell starts at zero and an observed transition ends the episode with reward r, that observation changes the cell to αr. Repeated experience refines this estimate and propagates information about rewards to earlier decisions.

The learning loop is

1. Initialise the table and reset the environment to its initial state.
2. Choose a random allowed action with probability ε, and otherwise choose an action maximising the current $Q(s,a)$.
3. Take the action and observe the next state, reward and terminal flag.
4. Update the visited cell using equation (3.3), with zero continuation at a terminal state.
5. Continue from the new state, or begin another episode after termination.

Exploration supplies information about alternative actions. Q-learning converges to the optimal action values under the usual tabular assumptions, which include sufficient visits to all relevant pairs and suitably decreasing learning rates [4]. A neural Q-network lacks these guarantees.

3.2 Double Q-learning

The largest of several noisy estimates tends to be too large, because the maximisation favours positive errors. Ordinary Q-learning uses the same table to choose the best next action and to evaluate it. Double Q-learning maintains two tables. When updating the first,

$$a^* = \arg\max_{a'} Q^A(s',a'), \tag{3.4}$$

$$y = r + \gamma Q^B(s',a^*), \tag{3.5}$$

$$Q^A(s,a) \leftarrow Q^A(s,a) + \alpha[y - Q^A(s,a)]. \tag{3.6}$$

The first table selects and the second evaluates, with the roles exchanged on other updates. Separating these operations reduces the maximisation bias [9].

4 Deep Q-learning and Double DQN

A deep Q-network (DQN) replaces the table by $Q(s,a;\theta)$. Its inputs are state features and its outputs are estimates of the available action values. The network is trained by regression,

$$\mathcal{L}(\theta) = \frac{1}{B}\sum_{i=1}^{B}[y_i - Q(s_i,a_i;\theta)]^2. \tag{4.1}$$

The target is held fixed while taking the gradient of this loss. Replay memory stores past transitions (s, a, r, s', d), where $d = 1$ indicates a terminal next state. Random minibatches reuse the observations and reduce the correlation between consecutive training samples. A target network with parameters θ^- is updated more slowly than the online network. DQN uses

$$y_i = r_i + \gamma(1 - d_i) \max_{a'} Q(s'_i, a'; \theta^-). \tag{4.2}$$

Replay and a target network make the regression more stable, although they give no general convergence guarantee [5].

Double DQN separates selection and evaluation,

$$a^*_i = \arg\max_{a'} Q(s'_i, a'; \theta), \qquad y_i = r_i + \gamma(1 - d_i) Q(s'_i, a^*_i; \theta^-). \tag{4.3}$$

The online network chooses the next action and the target network scores it, the neural-network version of the separation used in Double Q-learning [10]. A computational cutoff introduced only to limit sampling differs from termination of the task. Bootstrapping at such a truncation depends on the task definition, which the implementation must state.

5 Learning the policy directly

A policy network represents $\pi_\theta(a \mid s)$. With discrete actions it outputs preferences $z_\theta(s, a)$ and uses a softmax,

$$\pi_\theta(a \mid s) = \frac{e^{z_\theta(s,a)}}{\sum_{a'} e^{z_\theta(s,a')}}. \tag{5.1}$$

Training aims to increase the expected discounted return $\mathcal{J}(\theta) = \mathbb{E}_\theta[G(\tau)]$ of a trajectory τ. If the initial distribution and the environment dynamics do not depend on θ, its probability factors as

$$p_\theta(\tau) = p(s_0) \prod_t \pi_\theta(a_t \mid s_t) p(s_{t+1}, r_{t+1} \mid s_t, a_t). \tag{5.2}$$

For a differentiable policy with appropriate support we use $\nabla p = p \nabla \ln p$. The environment factors then disappear from the logarithmic derivative,

$$\nabla_\theta \mathcal{J} = \mathbb{E}_\theta[G(\tau) \nabla_\theta \ln p_\theta(\tau)] \tag{5.3}$$

$$= \mathbb{E}_\theta\Big[G(\tau) \sum_t \nabla_\theta \ln \pi_\theta(a_t \mid s_t)\Big]. \tag{5.4}$$

The method samples the environment without differentiating through its dynamics. Rewards received before an action cannot depend on that action, so their score-weighted expectation vanishes, giving the reward-to-go form

$$\nabla_\theta \mathcal{J} = \mathbb{E}_\theta\Big[\sum_t \gamma^t G_t \nabla_\theta \ln \pi_\theta(a_t \mid s_t)\Big]. \tag{5.5}$$

REINFORCE estimates this average from sampled episodes and takes a gradient-ascent step. On average, actions associated with larger returns become more probable [6]. A state-dependent baseline $b(s_t)$ can replace G_t by $G_t - b(s_t)$ and reduce variance. It leaves the mean gradient unchanged when it is independent of the sampled action, since $\sum_a \pi(a \mid s) \nabla \ln \pi(a \mid s) = 0$. A baseline estimated from the same samples requires care to avoid self-dependence.

An actor–critic method learns both the policy and a value estimate V_ϕ. The critic supplies the one-step signal

$$\delta_t = r_{t+1} + \gamma V_\phi(s_{t+1}) - V_\phi(s_t), \tag{5.6}$$

which estimates the advantage when the critic is accurate. The actor uses it to adjust its action probabilities, and the critic learns from the same reward and next-state target. Proximal policy optimisation (PPO), a useful extension, often uses this architecture together with a clipped probability-ratio objective that limits incentives for large policy changes. The compilation studies discussed here use DQN, Double DQN and REINFORCE.

5.1 Continuous angles

For a continuous action a, the policy may be Gaussian,

$$a \sim \mathcal{N}(\mu_\theta(s), \sigma_\theta(s)^2). \tag{5.7}$$

The network learns the mean and a positive width.

6 State preparation and the available operations

We know an initial state $|\psi_0\rangle$ and a target $|\psi_{\text{target}}\rangle$. A sequence of operations prepares

$$|\psi_m\rangle = \hat{U}|\psi_0\rangle, \qquad \hat{U} = \hat{U}_{a_{m-1}} \cdots \hat{U}_{a_1} \hat{U}_{a_0}, \qquad F = |\langle \psi_{\text{target}} | \psi_m \rangle|^2. \tag{6.1}$$

The fidelity F is between zero and one and equals one when the states agree up to a global phase. We maximise it subject to the available operations and a gate or time budget. Each action selects a gate from a specified set, or an evolution

$$\hat{U}_a = e^{-i\hat{\mathcal{H}}_a \Delta t_a}, \qquad \hbar = 1, \tag{6.2}$$

under an allowed Hamiltonian generator $\hat{\mathcal{H}}_a$ for an allowed duration Δt_a. The rightmost operation acts first. A physical target can be fixed even when the available operations reach it only approximately.

6.1 Primitive gates and their action

A gate acts on the complex amplitudes of the state. For one qubit, $|0\rangle = (1,0)^\top$, $|1\rangle = (0,1)^\top$ and $|\psi\rangle = a|0\rangle + b|1\rangle$, with $|a|^2 + |b|^2 = 1$. The transformed vector is $\hat{U}|\psi\rangle$, where $\hat{U}^\dagger \hat{U} = \hat{I}$. The Pauli matrices are

$$\hat{X} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \hat{Y} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \hat{Z} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The Hadamard and phase gates are

$$\hat{H} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad \hat{S} = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}, \quad \hat{T} = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}.$$

Hadamard maps $|0\rangle$ to $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ and $|1\rangle$ to $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$. The phase gates preserve the two computational-basis probabilities and change the phase between the amplitudes. For example, $\hat{T}|+\rangle = (|0\rangle + e^{i\pi/4}|1\rangle)/\sqrt{2}$. A relative phase can affect a later measurement after another gate causes interference.

With $c = \cos(\theta/2)$ and $s = \sin(\theta/2)$, rotations have the matrices

$$\hat{R}_x(\theta) = \begin{pmatrix} c & -is \\ -is & c \end{pmatrix}, \quad \hat{R}_y(\theta) = \begin{pmatrix} c & -s \\ s & c \end{pmatrix}, \quad \hat{R}_z(\theta) = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}.$$

They obey $\hat{R}_\alpha(\theta) = \exp[-i\theta\hat{\sigma}_\alpha/2]$. A Hamiltonian $\hat{\mathcal{H}} = (\Omega/2)\hat{\sigma}_\alpha$ applied for a duration Δt implements the angle $\theta = \Omega\Delta t$. The factor $1/2$ in the exponent makes θ the physical rotation angle. Both $\hat{R}_x(\theta)|0\rangle = c|0\rangle - is|1\rangle$ and $\hat{R}_y(\theta)|0\rangle = c|0\rangle + s|1\rangle$ give probability $\sin^2(\theta/2)$ of finding one, with different phases. A z rotation preserves the populations and changes their relative phase by θ. The phase gate satisfies $\hat{T} = e^{i\pi/8}\hat{R}_z(\pi/4)$.

The controlled-NOT (CNOT) gate has a control and a target. In the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$, with control first,

$$\widehat{\text{CNOT}}_{0\to1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

It preserves $|00\rangle$ and $|01\rangle$ and exchanges $|10\rangle$ with $|11\rangle$. By linearity this action applies to each component of a superposition. After $\hat{H}$ on the first qubit and CNOT on $|00\rangle$ the result is the entangled state $(|00\rangle + |11\rangle)/\sqrt{2}$. It cannot be expressed as the product of a state of the first qubit and a state of the second.

6.2 Gates as Hamiltonian evolutions

The Hadamard gate satisfies $\hat{H} = (\hat{X} + \hat{Z})/\sqrt{2}$ and $\hat{H}^2 = \hat{I}$. Its exact exponential forms are

$$\hat{H} = i \exp\left[-\frac{i\pi}{2\sqrt{2}}(\hat{X} + \hat{Z})\right] = \exp\left[-\frac{i\pi}{2}\left(\hat{I} - \frac{\hat{X} + \hat{Z}}{\sqrt{2}}\right)\right].$$

The first exponential is a rotation through π about the axis halfway between x and z, and the prefactor i is its global phase. The identity term in the second form moves that phase into the exponential. A generator $\hat{\mathcal{H}}_H = \pi(\hat{I} - \hat{H})/(2\Delta t)$ implements the exact matrix in duration Δt.

The control projectors $\hat{P}_0 = (\hat{I} + \hat{Z})/2$ and $\hat{P}_1 = (\hat{I} - \hat{Z})/2$ give $\widehat{\text{CNOT}} = \hat{P}_0 \otimes \hat{I} + \hat{P}_1 \otimes \hat{X}$. An exact exponential representation is

$$\widehat{\text{CNOT}} = \exp\left[-\frac{i\pi}{4}(\hat{I} - \hat{Z}) \otimes (\hat{I} - \hat{X})\right] = \exp\left[-\frac{i\pi}{4}(\hat{I}_4 - \hat{Z} \otimes \hat{I} - \hat{I} \otimes \hat{X} + \hat{Z} \otimes \hat{X})\right].$$

The generator vanishes for control zero and acts on the target as $\pi(\hat{I} - \hat{X})/2$ for control one, whose exponential is $\hat{X}$. The four terms of the expanded generator commute, and the expression gives the full CNOT matrix with all its phases.

6.3 Two-qubit rotations

For $\alpha = x, y, z$, we define $\hat{R}_{\alpha\alpha}(\theta) = \exp[-i\theta\hat{\sigma}_\alpha \otimes \hat{\sigma}_\alpha/2]$. Since the Pauli product squares to $\hat{I}_4$, it equals $c\hat{I}_4 - is\hat{\sigma}_\alpha \otimes \hat{\sigma}_\alpha$. In the same ordered two-qubit basis,

$$\hat{R}_{xx}(\theta) = \begin{pmatrix} c & 0 & 0 & -is \\ 0 & c & -is & 0 \\ 0 & -is & c & 0 \\ -is & 0 & 0 & c \end{pmatrix}, \quad \hat{R}_{yy}(\theta) = \begin{pmatrix} c & 0 & 0 & is \\ 0 & c & -is & 0 \\ 0 & -is & c & 0 \\ is & 0 & 0 & c \end{pmatrix},$$

$$\hat{R}_{zz}(\theta) = \begin{pmatrix} e^{-i\theta/2} & 0 & 0 & 0 \\ 0 & e^{i\theta/2} & 0 & 0 \\ 0 & 0 & e^{i\theta/2} & 0 \\ 0 & 0 & 0 & e^{-i\theta/2} \end{pmatrix}.$$

The XX and YY rotations mix $|00\rangle$ with $|11\rangle$ and $|01\rangle$ with $|10\rangle$. For example, $\hat{R}_{xx}(\theta)|00\rangle = c|00\rangle - is|11\rangle$, and YY gives the opposite sign of the imaginary amplitude. At $\theta = \pi/2$ these states are maximally entangled. The ZZ rotation preserves computational-basis probabilities and assigns phase $e^{-i\theta/2}$ to even parity and $e^{i\theta/2}$ to odd parity. Acting on $|++\rangle$ it can create entanglement through the relative phase. For each of these interactions, a Hamiltonian $(\Omega/2)\hat{\sigma}_\alpha \otimes \hat{\sigma}_\alpha$ implements $\theta = \Omega\Delta t$.

6.4 GHZ, W and Dicke targets

Each qubit has computational basis states $|0\rangle$ and $|1\rangle$. For N qubits the basis strings have length N, and a state has 2^N complex amplitudes. The Greenberger–Horne–Zeilinger (GHZ) state and the W state are

$$|\text{GHZ}_N\rangle = \frac{|0\ldots0\rangle + |1\ldots1\rangle}{\sqrt{2}}, \qquad |\text{W}_N\rangle = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} |0\ldots1_j\ldots0\rangle. \tag{6.3}$$

GHZ coherently combines the two aligned strings, and measuring all qubits gives all zeros or all ones with equal probability. The W state shares one excitation coherently among the N positions, so a measurement finds the excitation at each position with probability $1/N$. A coherent superposition has definite relative phases between its amplitudes. An incoherent mixture of the displayed strings has different quantum properties.

Both states are invariant under permutations of qubits, and both are entangled across every nonempty bipartition. Losing one qubit removes the coherence of a GHZ state from the remaining register and leaves an equal mixture of $|0\ldots0\rangle$ and $|1\ldots1\rangle$, which is separable. For the W state the remaining density matrix is

$$\hat{\rho}_{N-1} = \frac{N-1}{N}|\mathrm{W}_{N-1}\rangle\langle\mathrm{W}_{N-1}| + \frac{1}{N}|0\ldots0\rangle\langle0\ldots0|. \tag{6.4}$$

It retains entanglement for $N \geq 3$. The two families therefore test different forms of entanglement [3].

Dicke states generalise W to k shared excitations,

$$|D_N^{(k)}\rangle = \frac{1}{\sqrt{\binom{N}{k}}} \sum_{|x|=k} |x\rangle, \tag{6.5}$$

where $|x|$ counts the ones in a string. A small Dicke example makes the equal-amplitude rule explicit,

$$|D_4^{(2)}\rangle = \frac{|1100\rangle + |1010\rangle + |1001\rangle + |0110\rangle + |0101\rangle + |0011\rangle}{\sqrt{6}}.$$

Every component contains two excitations, and each placement has probability $1/6$ when all four bits are measured.

Our additional reference target is $|D_6^{(3)}\rangle$, with 20 equally weighted strings. It has a fixed excitation number, permutation symmetry and entanglement across every nonempty bipartition. W is $|D_N^{(1)}\rangle$. No trained-agent result is claimed for the three-excitation target.

6.5 Reference constructions

The Hadamard gate maps $|0\rangle$ to $(|0\rangle + |1\rangle)/\sqrt{2}$. A controlled-NOT (CNOT) flips its target qubit when its control is one. Starting from $|000000\rangle$, we apply $\hat{H}$ to qubit 0 and then $\widehat{\mathrm{CNOT}}_{0\to1}$, $\widehat{\mathrm{CNOT}}_{1\to2}$, $\widehat{\mathrm{CNOT}}_{2\to3}$, $\widehat{\mathrm{CNOT}}_{3\to4}$ and $\widehat{\mathrm{CNOT}}_{4\to5}$. The sequence produces $|\mathrm{GHZ}_6\rangle$ with unit fidelity, and the chain circuit uses N gates for N qubits, including $N-1$ CNOTs. With single-qubit gates and pairwise CNOTs, a GHZ circuit needs at least $N-1$ two-qubit connections to connect all qubits and at least one gate that creates a superposition. The chain construction is therefore shortest for that gate set.

An exact continuous construction of $|\mathrm{W}_6\rangle$ starts with $\hat{X}_0|000000\rangle = |100000\rangle$. On successive bonds $(j, j+1)$ we apply an excitation-preserving gate

$$\hat{G}(\theta)|10\rangle = \cos\theta|10\rangle + \sin\theta|01\rangle, \tag{6.6}$$

$$\hat{G}(\theta)|01\rangle = -\sin\theta|10\rangle + \cos\theta|01\rangle, \tag{6.7}$$

which leaves $|00\rangle$ and $|11\rangle$ unchanged. With $\cos\theta_j = 1/\sqrt{6-j}$ for $j = 0, \ldots, 4$, each operation leaves amplitude $1/\sqrt{6}$ on its left qubit and passes the remainder to the right, so after five such operations the six amplitudes are equal. These continuously chosen gates define a reference construction. A compiler restricted to $\hat{H}, \hat{S}, \hat{T}, \widehat{\mathrm{CNOT}}$ approximates this six-qubit target.

6.6 State preparation as a decision process

The prepared state $|\psi_t\rangle$ supplies the observation in the simulator. An action chooses a gate or a Hamiltonian evolution, and the transition applies that operation to the current state. The target is fixed for a training run, and the reward can be $r_{t+1} = F_{t+1} - F_t - \zeta_{a_t}$, with ζ_a a gate cost. The

return then rewards fidelity improvement and penalises expensive circuits. An episode begins at $|\psi_0\rangle$ and ends at success or at the specified task limit. The remaining budget belongs in the state when it changes the available continuation.

In these compilation simulations the network reads the real and imaginary parts of the exact amplitudes. A laboratory controller receives only measurements and may need an observation history or a state estimate. The cavity example develops that distinction.

6.7 Continuous compilation actions

In the continuous compilation task a layer chooses one angle per qubit and a discrete entangling operation on each bond. The allowed operations are CNOT, controlled-Z or no entangling gate. The final reward is the fidelity after the last layer. A model-based reference differentiates the fidelity through the simulator. The two approaches use different information, so their comparison must state these assumptions.

7 Compilation targets and results

The gate-compilation study uses $\hat{H}, \hat{S}, \hat{T}$ on individual qubits and nearest-neighbour CNOTs. The target is specified separately for every training run. Besides GHZ and W, the targets are ground states of an open Ising chain,

$$\hat{\mathcal{H}} = -J \sum_{i=0}^{N-2} \hat{Z}_i \hat{Z}_{i+1} - h \sum_{i=0}^{N-1} \hat{X}_i - g \sum_{i=0}^{N-1} \hat{Z}_i. \tag{7.1}$$

Here $\hat{X}_i$ and $\hat{Z}_i$ are Pauli operators on spin i, J is the coupling, h the transverse field and g the longitudinal field. A ground state is the eigenstate of lowest energy. For $g = 0$, $J = 1$ the transverse-field Ising model is integrable, and the compilation targets use $h = 0.5, 1, 2$. A second choice has an antiferromagnetic coupling and a longitudinal field, $J = -1$, $h = -0.9045$, $g = -0.8090$ in the gate-compilation study. With both fields nonzero the chain is not integrable, and its ground state is computed numerically. The equation and its parameters define the target.

We state every circuit together with its target and fidelity. In the recorded seed-0 run, a Double DQN agent found the four-gate circuit $\hat{H}_3$, $\widehat{\mathrm{CNOT}}_{3\to2}$, $\widehat{\mathrm{CNOT}}_{2\to1}$, $\widehat{\mathrm{CNOT}}_{1\to0}$ from $|0000\rangle$. Its target is $(|0000\rangle + |1111\rangle)/\sqrt{2}$, and its final fidelity is one. It is a shortest circuit for the specified gate set. This statement concerns this run and this target family only. Other runs may fail, and circuits for other target families need not be optimal.

The compilation results report separate target families, random seeds, gate budgets and achieved fidelities. Their comparisons show the behaviour of the particular agents within that budget. An agent may fail because the gate set approximates a target poorly, because exploration misses a useful sequence, or because optimisation is ineffective. Reference circuits and exhaustive searches on small instances help distinguish these cases. The Dicke target has an exact reference vector and no trained-agent comparison in these results.

8 Preparation with Hamiltonian evolutions

The same task can use evolutions under Hamiltonian terms in place of digital gates. The action selects a generator and an allowed duration, and the simulator applies its exponential to the whole spin chain. The target remains the ground state of a specified Hamiltonian. The agent learns a protocol, a sequence of generator and duration choices.

The analog study follows the two-resolution control scheme of Ref. [7]. A coarse agent first approaches the target, and a finer agent continues from a higher fidelity. The analog and gate studies use the same transverse-field Ising targets. Their longitudinal-field targets differ by sign choices that

rotate the ground state, so their fidelities do not describe the preparation of the same vector. A claim that one method performs better requires matched targets and resources.

9 One photon in a leaky cavity

A cavity mode is a quantum harmonic oscillator with photon-number states $|n\rangle$. The annihilation operator obeys $\hat{a}|n\rangle = \sqrt{n}|n-1\rangle$ and the number operator is $\hat{n} = \hat{a}^\dagger \hat{a}$. Photons escape at rate κ, and a classical drive with amplitude ε_t acts through

$$\hat{\mathcal{H}}_{\mathrm{d}}(t) = i\varepsilon_t(\hat{a}^\dagger - \hat{a}). \tag{9.1}$$

The goal is to prepare and maintain $|1\rangle$ despite this loss. A drive without photon-number feedback produces coherent states, whose number distribution is Poissonian and whose largest possible one-photon probability is $e^{-1} \approx 0.37$, reached at mean photon number one.

The source example of Ref. [8] adds continuous weak photon-number measurement. A recent segment of the noisy measurement trace is the observation supplied to the policy network. The policy chooses the next displacement drive, and training rewards the one-photon population $P_1 = \langle 1|\hat{\rho}|1\rangle$, where the density matrix $\hat{\rho}$ describes the cavity state. Photon-number measurement supplies information and changes the conditional state, which allows feedback to go beyond coherent driving alone. A quantum nondemolition measurement reads the photon number without absorbing the photons, which distinguishes it from detecting the emitted light.

In the feedback task the target is $\hat{\rho}_{\mathrm{target}} = |1\rangle\langle 1|$. The policy receives recent noisy photon-number readings and chooses the drive amplitude ε_t. We assign the immediate reward $r_{t+1} = P_1(t+1)$ and maximise $J(\theta) = \mathbb{E}_{\pi_\theta}[T^{-1}\sum_{t=0}^{T-1} r_{t+1}]$, the expected one-photon population averaged over the episode. This objective rewards maintaining the state despite photon loss. The reward is computed from the simulated state during training. An experimental implementation needs a state estimate or repeated photon-number tests to evaluate it. The noisy observation and the reward therefore provide different information. A mean photon number of one alone does not establish that the cavity is in $|1\rangle$.

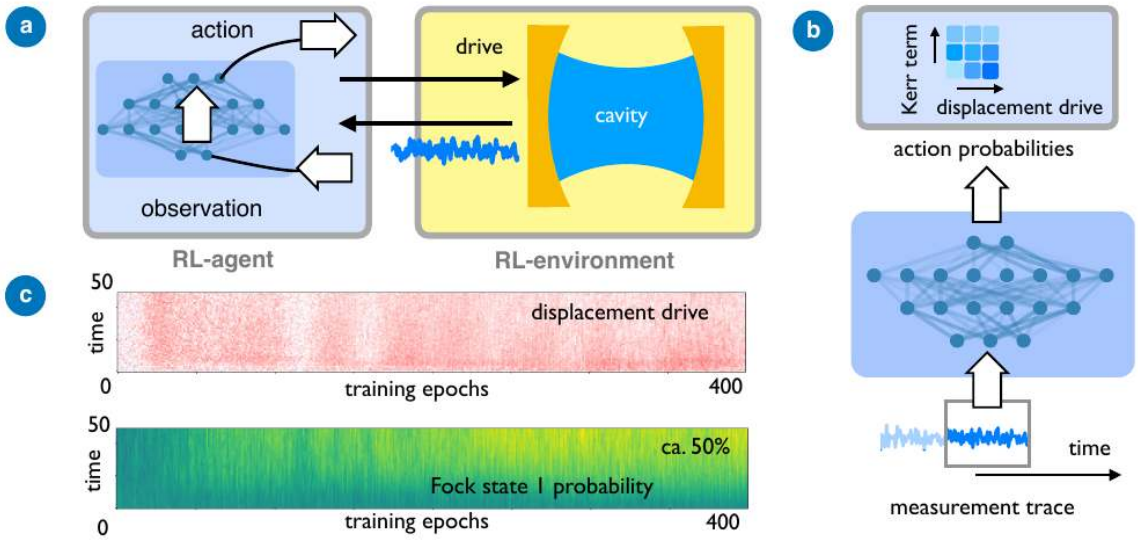


Figure 1: Source figure of the cavity feedback study. Panel (a) shows the control loop, panel (b) maps a recent measurement trace to action probabilities, and panel (c) shows the drive and one-photon probability during training. The one-photon probability approaches about 50% in this example. The figure is a source result, separate from our simulation.

Our simulation retains one lossy mode, discrete drive choices, a window of recent observations and policy-gradient learning. It simplifies the meter to discrete readings $x_t = n + \sigma_{\mathrm{m}}\xi_t$, where ξ_t is a unit Gaussian random variable, and updates the conditional state using the measurement likelihood. Its

numerical parameters and trained results belong to this simulation alone. The source figure motivates the feedback problem and shows its goal, and our simplified simulation is a separate study.

9.1 Open evolution and measurement

For a prescribed drive and weak photon-number measurement, a useful convention for the average state is

$$\frac{d\hat{\rho}}{dt} = -i[\hat{\mathcal{H}}_{\mathrm{d}}, \hat{\rho}] + \kappa \mathcal{D}[\hat{a}]\hat{\rho} + \Gamma_{\mathrm{m}} \mathcal{D}[\hat{n}]\hat{\rho}, \qquad \mathcal{D}[\hat{L}]\hat{\rho} = \hat{L}\hat{\rho}\hat{L}^{\dagger} - \tfrac{1}{2}\{\hat{L}^{\dagger}\hat{L}, \hat{\rho}\}.$$

The three terms describe the drive, photon loss and measurement dephasing. For ideal measurement the conditional state and measurement record obey

$$\begin{aligned} d\hat{\rho}_{\mathrm{c}} &= \big(-i[\hat{\mathcal{H}}_{\mathrm{d}}, \hat{\rho}_{\mathrm{c}}] + \kappa \mathcal{D}[\hat{a}]\hat{\rho}_{\mathrm{c}} + \Gamma_{\mathrm{m}} \mathcal{D}[\hat{n}]\hat{\rho}_{\mathrm{c}} \big) dt + \sqrt{\Gamma_{\mathrm{m}}}\, \mathcal{M}[\hat{n}]\hat{\rho}_{\mathrm{c}}\, dW, \\ dY &= 2\sqrt{\Gamma_{\mathrm{m}}}\, \mathrm{Tr}(\hat{n}\hat{\rho}_{\mathrm{c}}) dt + dW, \\ \mathcal{M}[\hat{n}]\hat{\rho}_{\mathrm{c}} &= \hat{n}\hat{\rho}_{\mathrm{c}} + \hat{\rho}_{\mathrm{c}}\hat{n} - 2\,\mathrm{Tr}(\hat{n}\hat{\rho}_{\mathrm{c}})\hat{\rho}_{\mathrm{c}}. \end{aligned}$$

Here Γ_{m} is the measurement strength and dW is a Wiener increment of mean zero and variance dt. The stochastic term updates the state conditioned on the noisy record. A feedback policy makes the drive depend on the past record. The unconditional equation with prescribed drive therefore cannot be used to factor an average of a feedback-dependent drive and a conditional state.

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